Minor Project Report: Data Cleaning & Quality Assessment

Student Name: Deepak Kumar Sahoo

b **Dataset Used:** Healthcare Dataset (Medical Cost Personal Dataset) **Project Link:** GitHub Repository

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1. Project Overview

Real-world datasets often contain missing, inconsistent, and noisy data. This project focuses on cleaning and preparing a raw insurance/healthcare dataset for downstream analysis and predictive modeling. The primary objectives were to identify missing, duplicate, and inconsistent values, apply appropriate cleaning techniques (such as imputation and encoding alignment), handle statistical outliers systematically, and document all data quality issues and resolutions.

2. Tools & Technologies Used

- **Python:** Primary programming language used for scripting the data pipeline.
- **Pandas:** Utilized for data ingestion, duplicate detection, boolean indexing, and dataset exportation.
- **NumPy:** Used to support statistical boundary calculations.
- **Jupyter Notebook:** Interactive development environment for executing data preprocessing steps.

3. Data Quality Issues & Fixes Applied

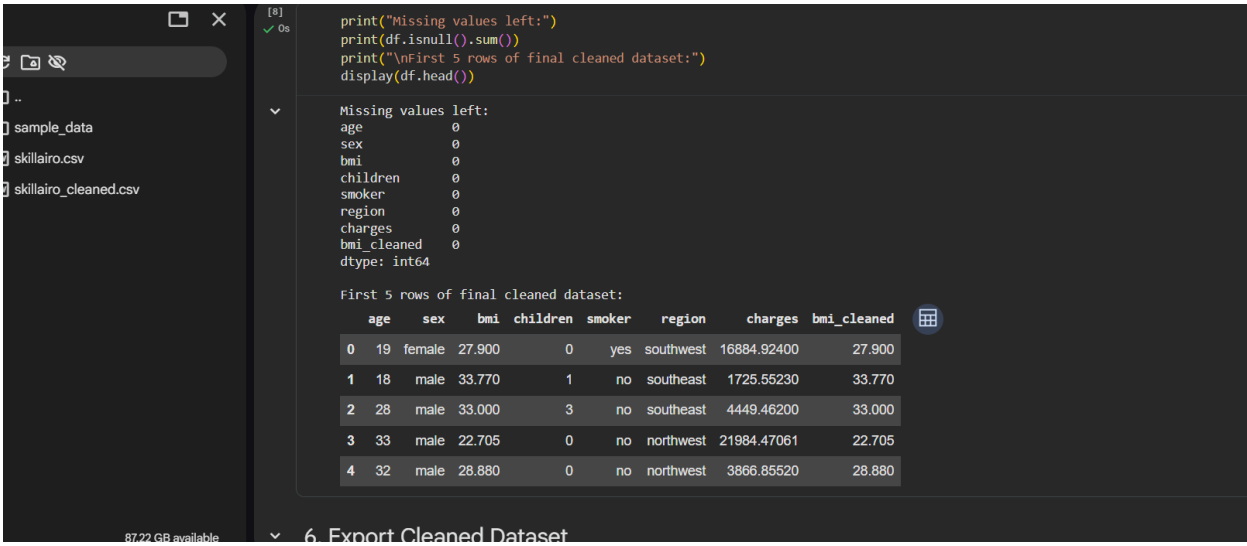
A systematic approach was taken to clean the raw data (1,338 original rows):

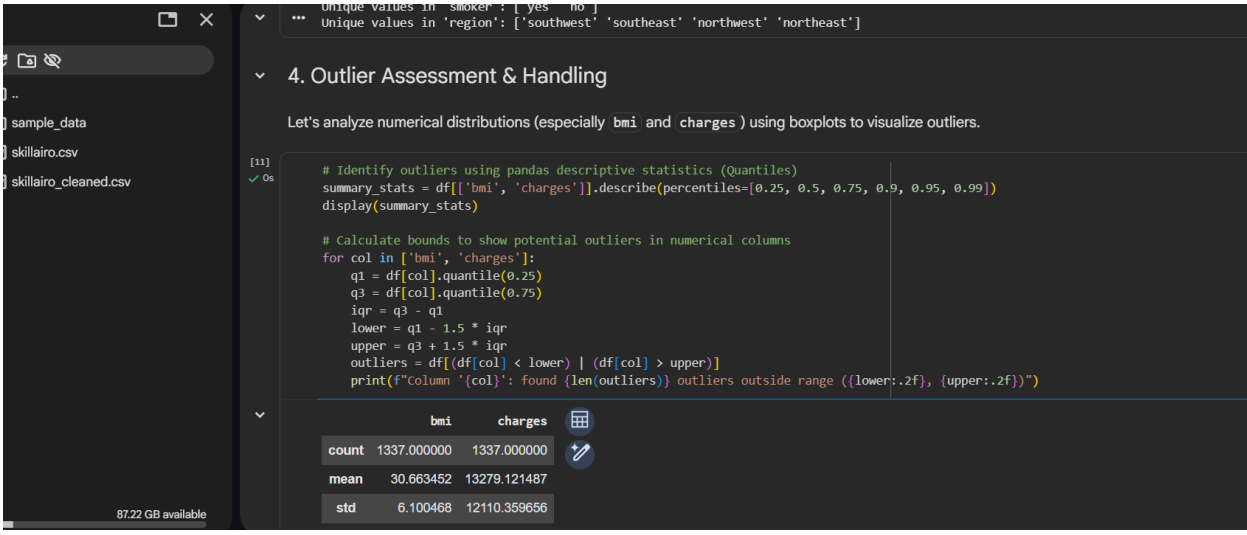
- **Missing Values Assessment:**
    - *Issue:* Real-world data frequently contains null values that break machine learning algorithms.

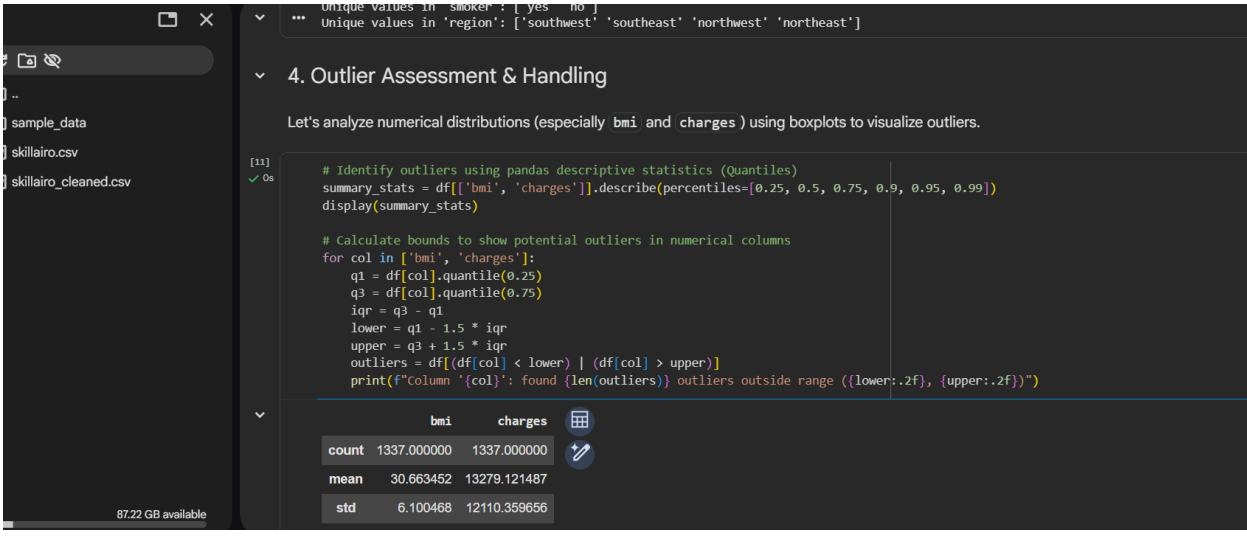
    - *Fix Applied:* Code was implemented to check for nulls. After cleaning, it was confirmed that zero missing values remained across all columns (age, sex, bmi, children, smoker, region, charges).
- **Inconsistent Data (Duplicates & Formatting):**
    - *Issue:* Duplicate records and inconsistent text casing can skew data weightings.
    - *Fix Applied:* Text variables (sex, smoker) were harmonized to lowercase. The data quality assessment identified 1 exact duplicate entry. The `.drop_duplicates()` function was applied to remove the redundant record.
- **Handling Outliers (BMI & Charges):**
    - *Issue:* Healthcare data is highly sensitive to extreme values.
    - *Fix Applied:* The Interquartile Range (IQR) method was applied to numerical distributions using descriptive statistics to calculate upper and lower bounds. A new column named `bmi_cleaned` was engineered to "cap" (Winsorize) the extreme outliers at the upper statistical boundary (47.315). This preserved the dataset's structural integrity without deleting valuable patient records.

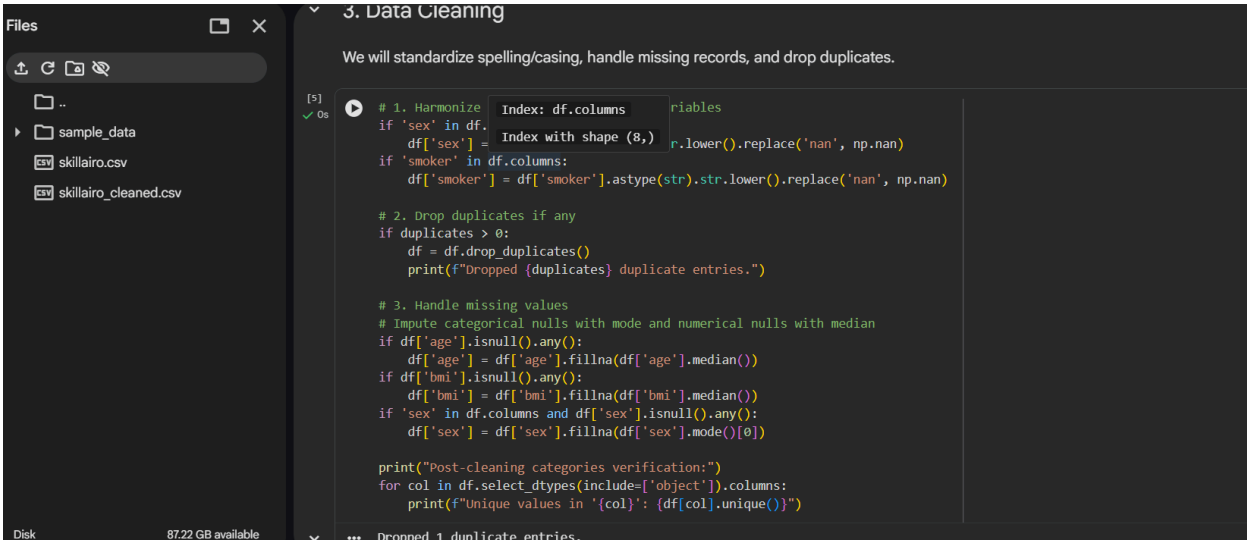
4. Final Outcome

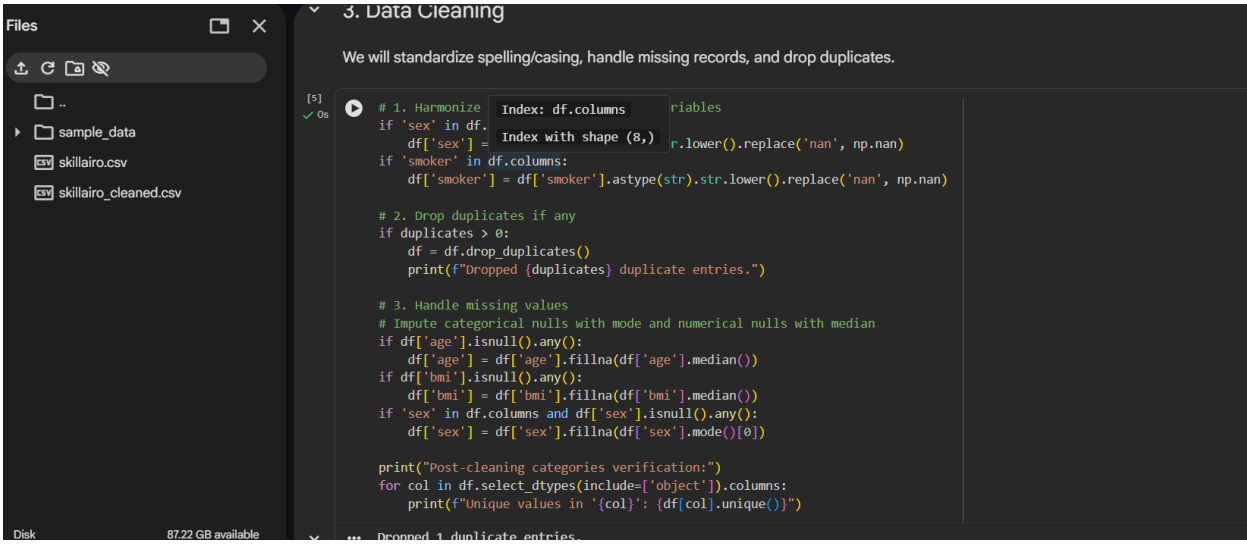
- The dataset was successfully processed and refined. The row count was reduced to 1,337 cleanly structured records, achieving a normalized BMI distribution without losing critical data. The final structured dataset was exported as `minorpro(finalone).xlsx`, establishing a strong foundation for future predictive modeling.


[8]
0s
print("Missing values left:")
print(df.isnull().sum())
print("\nFirst 5 rows of final cleaned dataset:")
display(df.head())
Missing values left:
age 0
sex 0
bmi 0
children 0
smoker 0
region 0
charges 0
bmi_cleaned 0
dtype: int64
First 5 rows of final cleaned dataset:
age sex bmi children smoker region charges bmi_cleaned
0 19 female 27.900 0 yes southwest 16884.92400 27.900
1 18 male 33.770 1 no southeast 1725.55230 33.770
2 28 male 33.000 3 no southeast 4449.46200 33.000
3 33 male 22.705 0 no northwest 21984.47061 22.705
4 32 male 28.880 0 no northwest 3866.85520 28.880
sample_data
skillairo.csv
skillairo_cleaned.csv
87.22 GB available
6. Export Cleaned Dataset


Unique values in 'smoker': ['yes' 'no']
Unique values in 'region': ['southwest' 'southeast' 'northwest' 'northeast']
4. Outlier Assessment & Handling
Let's analyze numerical distributions (especially bmi and charges) using boxplots to visualize outliers.
[11]
0s
Identify outliers using pandas descriptive statistics (Quantiles)
summary_stats = df[['bmi', 'charges']].describe(percentiles=[0.25, 0.5, 0.75, 0.9, 0.95, 0.99])
display(summary_stats)
Calculate bounds to show potential outliers in numerical columns
for col in ['bmi', 'charges']:
q1 = df[col].quantile(0.25)
q3 = df[col].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
outliers = df[(df[col] < lower) | (df[col] > upper)]
print(f"Column '{col}': found {len(outliers)} outliers outside range ({lower:.2f}, {upper:.2f})")
bmi charges
count 1337.000000 1337.000000
mean 30.663452 13279.121487
std 6.100468 12110.359656
sample_data
skillairo.csv
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Files
..
sample_data
skillairo.csv
skillairo_cleaned.csv
Disk
87.22 GB available
3. Data Cleaning
We will standardize spelling/casing, handle missing records, and drop duplicates.
[5]
0s
Index: df.columns
Index with shape (8,)
1. Harmonize riables
if 'sex' in df.
df['sex'] = r.lower().replace('nan', np.nan)
if 'smoker' in df.columns:
df['smoker'] = df['smoker'].astype(str).str.lower().replace('nan', np.nan)
2. Drop duplicates if any
if duplicates > 0:
df = df.drop_duplicates()
print(f"Dropped {duplicates} duplicate entries.")
3. Handle missing values
Impute categorical nulls with mode and numerical nulls with median
if df['age'].isnull().any():
df['age'] = df['age'].fillna(df['age'].median())
if df['bmi'].isnull().any():
df['bmi'] = df['bmi'].fillna(df['bmi'].median())
if 'sex' in df.columns and df['sex'].isnull().any():
df['sex'] = df['sex'].fillna(df['sex'].mode()[0])
print("Post-cleaning categories verification:")
for col in df.select_dtypes(include=['object']).columns:
print(f"Unique values in '{col}': {df[col].unique()}")
Dropped 1 duplicate entries.


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```
print("=== Dataset Info ===")
df.info()
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go)
nt("\n=== Missing Values Per Column ===")
issing_info = df.isnull().sum()
display(missing_info[missing_info > 0])

# Duplicate rows
duplicates = df.duplicated().sum()
print(f"\nNumber of duplicate rows: {duplicates}")
```

```
=== Dataset Info ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB

=== Missing Values Per Column ===
```

```
    print("File not found. Creating a simulation of the insurance dataset for demonstration.")
    np.random.seed(42)
    n_samples = 1000
    df = pd.DataFrame({
        'age': np.random.choice([19, 28, 33, 45, 50, 64, None], size=n_samples, p=[0.15, 0.15, 0.20, 0.20, 0.15, 0.10, 0.05]),
        'sex': np.random.choice(['female', 'male', 'FEMALE', 'MALE', None], size=n_samples, p=[0.45, 0.45, 0.04, 0.04, 0.02]),
        'bmi': np.random.normal(loc=30, scale=6, size=n_samples),
        'children': np.random.choice([0, 1, 2, 3, 4, 5], size=n_samples),
        'smoker': np.random.choice(['yes', 'no', 'YES', 'NO'], size=n_samples, p=[0.18, 0.78, 0.02, 0.02]),
        'region': np.random.choice(['southwest', 'southeast', 'northwest', 'northeast'], size=n_samples),
        'charges': np.random.exponential(scale=13000, size=n_samples) + 1000
    })
    # Introduce some outliers and missin                ntionally
    df.loc[df.sample(frac=0.03).index, 'bmi'] = np.nan
    df.loc[df.sample(frac=0.01).index, 'bmi'] = 99.9  # extreme value outlier

display(df.head())
print(f"Dataset dimensions: {df.shape[0]} rows, {df.shape[1]} columns")
```

Loading...

```
Dataset loaded successfully!
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

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Problem Statement: Real-world datasets often contain missing, inconsistent, and noisy data. This notebook focuses on cleaning and preparing a raw insurance/healthcare dataset for downstream analysis and modeling.

Objective:

- Identify missing, duplicate, and inconsistent values.
- Apply appropriate cleaning techniques (imputation, encoding alignment).
- Handle outliers systematically.
- Document data quality issues and resolutions.

[10] ✓ 0s

```
import pandas as pd
import numpy as np

# Configure display options for better DataFrame printing
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 1000)
```

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